Learning Objective 9.5: I can determine the reflection and transmission of UPW at a boundary for Transverse Electric (TE) and Transverse Magnetic (TM) waves, using boundary conditions.

Show your work. Write a **Documentation** line at the end (**None** if you did not collaborate). In every problem the boundary is the plane $z = 0$, medium 1 is $z < 0$, the plane of incidence is the xz plane, and both materials are lossless and nonmagnetic, so $\eta = \eta_0/n$ with $\eta_0 = 376.7\,\Omega$.

1. **Phase matching.** A 10 GHz plane wave in air strikes a dielectric with $\epsilon_r = 4$ at $\theta_i = 60^\circ$.

   (a) The tangential electric field at $z = 0$ must satisfy

   $$E_0 e^{-j\beta_1 x \sin\theta_i} + \Gamma_{\mathsf{TE}} E_0 e^{-j\beta_1 x \sin\theta_r} = \tau_{\mathsf{TE}} E_0 e^{-j\beta_2 x \sin\theta_t}$$

   for every x. Explain why this requires the three exponents to be equal, and derive the law of reflection and Snell's law from it.

   (b) Find the transmitted angle.

   $\theta_t =$ ☐

   (c) Find the wavelength in each medium, and the spacing along the boundary between points where each wave has the same phase. Show that the two spacings agree.

   spacing = ☐

2. **The TE coefficients.** The wave of Question 1 is TE, with $\widetilde{\mathbf{E}}_i = \hat{\mathbf{y}}\, E_0\, e^{-j\beta_1(x\sin\theta_i + z\cos\theta_i)}$.

   (a) Write the reflected and transmitted electric and magnetic fields, using $\widetilde{\mathbf{H}} = \hat{\mathbf{k}} \times \widetilde{\mathbf{E}}/\eta$. Apply the two tangential boundary conditions and derive Γ_{TE} and τ_{TE}.

   (b) Evaluate Γ_{TE} and τ_{TE}.

   Γ_{TE} =

   τ_{TE} =

   (c) Find the fraction of the incident power that reflects. Use Lesson 39's transmitted fraction, $T = |\tau_{\text{TE}}|^2(\eta_1\cos\theta_t)/(\eta_2\cos\theta_i)$, to check that the reflected and transmitted fractions add to one.

   reflected =

3. **Limits and a second boundary.**

   (a) Show that Γ_{TE} reduces to Lesson 38's normal-incidence Γ when $\theta_i = 0$, and find its limit as $\theta_i \to 90^\circ$.

   (b) A TE wave in glass, $n_1 = 1.50$, meets water, $n_2 = 1.33$, at $\theta_i = 20^\circ$. Find θ_t, Γ_{TE}, and τ_{TE}.

   θ_t =

   Γ_{TE} =

   (c) Explain how τ_{TE} can be greater than one without the transmitted wave carrying more power than the incident wave.

Documentation: _____